

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 3035	Team ID	SWTID-2026-6491
Project Name	Import Data Using Transform Maps	Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2.

Example: Import Data Using Transform Maps

Reference: ServiceNow Import Sets and Transform Maps

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / Angular Js / React Js etc.
2	Application Logic-1	Logic for a process in the application	Java / Python
3	Application Logic-2	Logic for a process in the application	IBM Watson STT service
4	Application Logic-3	Logic for a process in the application	IBM Watson Assistant
5	Database	Data Type, Configurations etc.	MySQL, NoSQL, etc.
6	Cloud Database	Database Service on Cloud	IBM DB2, IBM Cloudant etc.
7	File Storage	File storage requirements	IBM Block Storage or Other Storage Service or Local Filesystem
8	External API-1	Purpose of External API used in the application	IBM Weather API, etc.
9	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.
11	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud	Local Server Configuration: Cloud Server Configuration : Local, Cloud Foundry, Kubernetes, etc.

Import Data Using Transform Maps – Topic Content

Import Set: Incoming source data is loaded into a staging table before transformation.

Transform Map: Defines how source fields are mapped to fields in the target ServiceNow table.

Field Mapping: Maps source fields to target fields and can apply transformation logic.

Coalesce: Helps identify existing records so imported data can update records instead of creating duplicates.

Transform Flow: Source Data → Import Set Table → Transform Map → Field Mapping / Transformation → Target Table.

Validation: Transformation logic can validate and standardize incoming data before it reaches the target table.

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	ServiceNow platform processing with JavaScript for custom transformation logic.	ServiceNow Platform / JavaScript
2	Security Implementations	Role-based access, ACL controls, authentication, authorization and encrypted communication.	ServiceNow ACLs / Roles / HTTPS
3	Scalable Architecture	Import-set staging and Transform Maps provide controlled processing before data reaches target tables.	ServiceNow Cloud / Import Sets / Transform Maps

4	Availability	Cloud-hosted ServiceNow platform provides managed application availability.	ServiceNow Cloud Infrastructure
5	Performance	Efficient field mapping, coalesce configuration and optimized transform scripts reduce unnecessary processing.	Transform Maps / Coalesce / JavaScript

References:

ServiceNow documentation – Import Sets and Transform Maps

C4 Model – <https://c4model.com/>

IBM Cloud Architecture – <https://www.ibm.com/cloud/architecture>